CHAPTER 21 · DERIVATIONS

WAVE MOTION

3
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PHASE DIFFERENCE

SYLLABUS 21.1(D)

RESULT $\varphi = kx = \dfrac{2\pi x}{\lambda}$

1. Two points one wavelength apart differ in phase by 2π.
2. Phase difference is proportional to separation: $\dfrac{\varphi}{2\pi} = \dfrac{x}{\lambda}$.
3. $\boldsymbol{\varphi = \dfrac{2\pi x}{\lambda} = kx}$, with $k = \dfrac{2\pi}{\lambda}$.

2 WAVE SPEED

SYLLABUS 21.1(E)

RESULT $v = f\lambda$

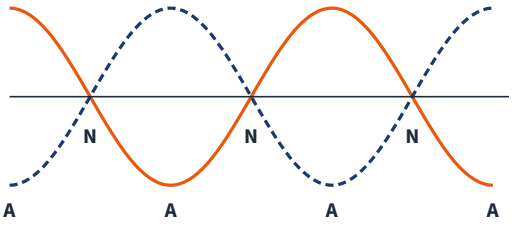
1. In one period T the wave moves forward one wavelength λ.
2. $v = \dfrac{\lambda}{T}$, and $f = \dfrac{1}{T}$, so $\boldsymbol{v = f\lambda}$ (equivalently $v = \dfrac{\omega}{k}$).

3 STANDING WAVE EQUATION

SYLLABUS 21.4(B)

RESULT $y = 2A \cos kx \sin \omega t$

1. Two identical waves travel in opposite directions: $y_1 = A \sin(\omega t - kx)$, $y_2 = A \sin(\omega t + kx)$.
2. Superposition: $y = y_1 + y_2$. Use $\sin P + \sin Q = 2 \sin \dfrac{P+Q}{2} \cos \dfrac{P-Q}{2}$.
3. $\boldsymbol{y = 2A \cos kx \sin \omega t}$.
4. Amplitude $2A \cos kx$ depends on position: nodes where $\cos kx = 0$ ($x = \dfrac{\lambda}{4}, \dfrac{3\lambda}{4}, \ldots$), antinodes where $|\cos kx| = 1$ ($x = 0, \dfrac{\lambda}{2}, \lambda, \ldots$). Adjacent nodes are $\dfrac{\lambda}{2}$ apart; no energy is carried along.



N = node, A = antinode; N to N = λ/2